

Augmented Lagrangian Predictive Coding

Jeffrey Seely

Sakana AI

jeffrey@sakana.ai

Julian Gould

Sakana AI

juliangould@sakana.ai

Abstract

Predictive coding (PC) is a local-learning alternative to backpropagation (BP), training deep networks via **local energy-minimization dynamics rather than a global backward pass**. We introduce Augmented Lagrangian Predictive Coding (PC-ALM), which maintains PC’s inference budget but aligns each weight update toward BP by accumulating per-layer constraint errors into a layer-local Lagrange multiplier. In linear PC networks, **PC-ALM converges to an equilibrium with exact BP gradients distributed across the network via only layer-local updates**. We analyze PC-ALM in nonlinear PC networks up to depth 128 and show that it matches BP performance across all width-depth regimes, notably in deep narrow networks where PC underperforms. PC-ALM introduces recurrent dynamics in each layer’s activations. Compared to PC’s heat flow on a scalar energy, PC-ALM dynamics are driven by dual ascent on the augmented Lagrangian. We observe “ballistic” credit propagation across very deep networks, with credit signals evenly distributed across layers, compared to PC’s slow, diffusive credit propagation. Beyond the algorithm itself, the augmented Lagrangian framework offers a generalization of PC, and may yield insights into how distributed systems could compute and propagate BP-like credit signals through purely local dynamics.

1 Introduction

Modern deep learning relies on backpropagation (BP). Yet biological circuits cannot implement BP, at least not exactly [Lillicrap et al., 2020]. One reason is that BP is a *global* algorithm: even though BP’s chain rule spatially localizes updates, the network must still create error variables in parallel with the matching forward activations, then gate weight updates in a precisely orchestrated sequence of steps. This gap has motivated research into *local* alternatives to BP, where state, error, and weight variables evolve as a single dynamical system that requires only layer-local coupling. Predictive coding (PC) is one of the more actively researched layer-local alternatives to BP [Whittington and Bogacz, 2017, Millidge et al., 2021, Salvatori et al., 2026]. In a Predictive Coding Network (PCN), a mean-squared error term is introduced between each layer, the sum of which is the PC energy. Gradient descent on this energy yields a layer-local dynamical system for updating the activation states (PC inference) and weights (PC learning). At fixed weights, the activation equilibrium yields a different activation pattern than one would get from a typical forward pass, resulting in a corresponding weight update that is aligned with, but not equal to, the BP gradient.

In this paper, we frame PC within a broader class of local learning methods involving the augmented Lagrangian (AL). Dual ascent on the AL—known as the method of multipliers (MM) [Hestenes, 1969]—yields a local algorithm in certain cases and has been extensively used in decentralized approaches to network training (e.g., Taylor et al. [2016]). We show that MM can be used as a replacement for PC’s inference step, but with a substantial difference: at equilibrium, the Lagrange multipliers converge to exactly the gradient signal of the supervised loss.

This connection is not entirely new. LeCun [1988] observed that the multipliers of the *standard* Lagrangian are the BP adjoints at feasibility. Interestingly, the augmented Lagrangian is essentially

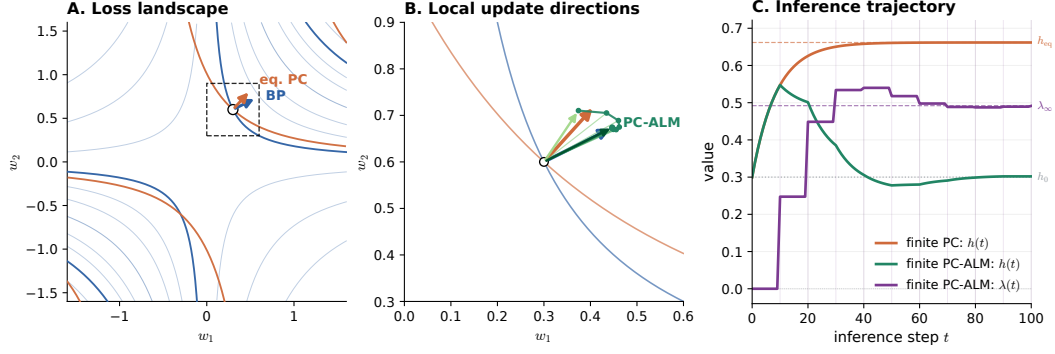


Figure 1: PC-ALM credit alignment in a scalar two-layer network ($y = w_2 w_1 x$; $h = w_1 x$; $x=y=1$). (A) BP loss landscape with BP and equilibrated-PC iso-loss curves. (B) Arrows are negative weight gradients for BP, equilibrated PC, and PC-ALM at different iterations. (C) Inference trace. PC settles h at h_{eq} ; PC-ALM returns h to $h_0 = w_1 x$ while λ converges to the BP adjoint $\lambda_\infty = (y - w_2 h_0) w_2$. Here, we take 10 primal steps before each dual step to emphasize the role of the dual as a lagged integrator of error; in Algorithm 1 we alternate between one primal and one dual step.

the standard Lagrangian plus the PC energy (up to the shared supervised-loss term), suggesting that there is a close relationship between PC and augmented Lagrangian methods.

Contribution: PC-ALM. The caveat of MM is that exact BP computation requires convergence to equilibrium in its inner solve, which is not guaranteed for general nonlinear networks [Wang et al., 2019]. However, we note that this difficulty is shared by PC. Nevertheless, the PC literature handles this empirically, fixing a finite inference budget and assessing performance directly [Innocenti et al., 2025, 2026, Pinchetti et al., 2025], often justified by using good initializations that are close to equilibrium. Motivated by this, we develop a finite-inference variant of MM, called PC-ALM (for Augmented Lagrangian Method), which operates on the same inference iteration budget as PC.

Mechanistically, PC-ALM modifies PC inference with a dual variable (i.e. Lagrange multiplier) per layer that integrates prediction errors and feeds back into the local activations. After a fixed number of inference steps, the weight update acts on a composite signal of the prediction error shifted by the dual. The recurrence introduces stable oscillatory transients on an intermediate time scale between PC’s fast inference and slow weight updates.

We first show that in linear PC networks, PC-ALM converges to a solution with exact BP gradients. Our core experimental finding is that PC-ALM closes the PC-BP gap at matched inference budget in nonlinear networks. Following Innocenti et al. [2026], we test PC-ALM at depths up to 128 on image classification. PC-ALM matches BP across all width-depth regimes, including the deep narrow networks where PC significantly underperforms BP.

We further observe experimentally that PC-ALM exhibits surprising dynamical properties. PC networks are known to concentrate credit signals near the supervised loss [Pinchetti et al., 2025]. In contrast, PC-ALM distributes credit near-uniformly across the network. Finally, we observe that the primal-dual dynamics of PC-ALM accelerate the credit-propagation wavefront in extremely deep networks, where standard PC stalls even at equilibrium.

2 Related Work

Our current work draws on an under-explored connection between the biologically plausible deep learning literature and the distributed optimization literature.

Predictive coding. In biologically plausible deep learning, one core research direction has been to make learning more local. Predictive coding originated in computational neuroscience as a model of cortical inference [Rao and Ballard, 1999, Friston and Kiebel, 2009] and has seen ongoing development [Millidge et al., 2021, Salvatori et al., 2026] as a layer-local approximation to BP

using only Hebbian plasticity. Millidge et al. [2022a] established a unified inference-and-learning framework, Salvatori et al. [2022] extended PC to arbitrary graph topologies, and Alonso et al. [2024] characterized the optimization difficulty of PC inference and proposed accelerated solvers. The current empirical state of the field is set by Pinchetti et al. [2025], who find that PC can match BP on small-to-medium image-classification tasks but lags substantially for deep networks. Innocenti et al. [2024] show that the equilibrated PC energy contains only strict saddles, and Innocenti et al. [2025] introduce a μ P-style parameterization (μ PC) which enables stable training for deep PC networks; follow-up work [Innocenti et al., 2026] characterizes the PC-BP gap as a function of network width and depth.

Augmented Lagrangian and ADMM for network training. A separate community has trained networks by alternating direction method of multipliers (ADMM) or augmented-Lagrangian methods, often for scalability, conditioning, or distributed optimization rather than biological plausibility. Carreira-Perpiñán and Wang [2014] introduce the method of auxiliary coordinates with a quadratic-penalty energy and note a connection to augmented-Lagrangian methods. Taylor et al. [2016] use a single output-layer Bregman/multiplier correction and report full per-constraint multipliers as unstable in their setup. Zeng et al. [2021] attach multipliers to each layer consistency constraint and prove $O(1/k)$ convergence to a KKT point under smooth-activation assumptions. Wang et al. [2025] write Elman RNN training with multipliers on both the affine and the nonlinear constraints, solve each block in closed form by exploiting ReLU’s piecewise-linear structure, and prove convergence to a KKT point; applied to a feedforward MLP, this is analogous to PC-ALM with separate affine and activation constraints/multipliers. Evens et al. [2021] formulate feedforward network training as a nonlinear optimal-control problem and apply ALM with Gauss–Newton and forward dynamic programming. Frerix et al. [2018] introduce ProxProp, a penalty-only formulation of network training in which a forward pass followed by a reverse-order sweep of gradient steps on a quadratic-penalty energy reproduces one BP step exactly. ProxProp carries no multipliers and runs no outer dual iteration. Lifted-training methods [Askari et al., 2018, Li et al., 2018, Gu et al., 2020, Zach and Estellers, 2019, Wang and Benning, 2023, Wang et al., 2026] replace hard layer constraints with activation-specific penalties or relaxations (proximal, Fenchel/Lagrange-relaxation, contrastive, Bregman), but generally do not carry persistent per-sample layer multipliers in the inference state. Zhou et al. [2026] run stochastic ADMM over data/client-partitioned local parameter copies with consensus multipliers, splitting on a different axis than the layer-local splitting used here. Where these methods carry multipliers, they typically update them after ADMM/ALM block rounds or local subproblem updates.

3 Background

We start by reviewing the constrained optimization formalism for training a deep feedforward network.

Constrained network. Consider a feedforward network of depth L with weights $\theta = \{W_i\}_{i=1}^L$ and elementwise nonlinearity σ (e.g., identity, tanh, ReLU). Given input $x \in \mathbb{R}^{n_0}$ and target $y \in \mathbb{R}^{n_L}$, the forward pass computes $h_i = \sigma(W_i h_{i-1})$ from $h_0 = x$ for $i = 1, \dots, L-1$, with linear readout $W_L h_{L-1}$. We can write the optimization problem as

$$\begin{aligned} \min_{\theta, h} \quad & \frac{1}{2} \|y - W_L h_{L-1}\|^2 \\ \text{s.t.} \quad & h_i = \sigma(W_i h_{i-1}), \quad i = 1, \dots, L-1, \end{aligned} \tag{1}$$

where the activations $h = \{h_i\}_{i=1}^{L-1}$ are optimization variables alongside the weights θ .

Predictive coding as quadratic penalty. The penalty relaxation of (1) replaces each layer constraint with a squared-error term, giving the PC energy [Whittington and Bogacz, 2017, Millidge et al., 2022a, Salvatori et al., 2022]:

$$\mathcal{F}_{\text{PC}}(h, \theta) = \frac{1}{2} \|y - W_L h_{L-1}\|^2 + \frac{\rho}{2} \sum_{i=1}^{L-1} \|h_i - \sigma(W_i h_{i-1})\|^2, \tag{2}$$

for a penalty strength ρ .

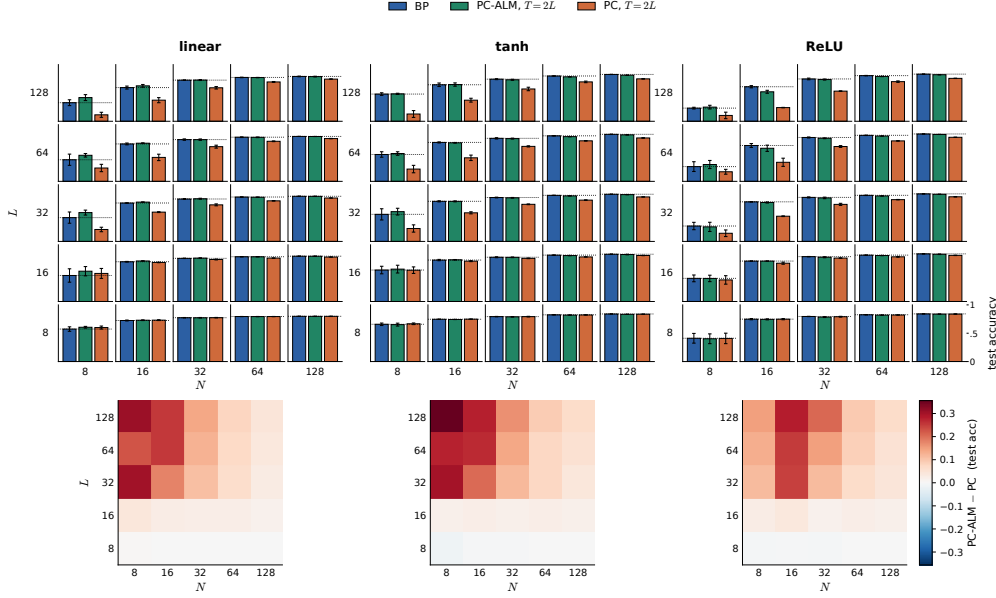


Figure 2: Width-depth results on Fashion-MNIST. PC-ALM matches BP at a budget of $T = 2L$ across the full range of N , L , and activations. PC-ALM outperforms PC in deep narrow networks.

Equilibrated PC. For each sample (x, y) , PC alternates between two steps:

$$\text{inference: } h^* = \arg \min_h \mathcal{F}_{\text{PC}}(h, \theta), \quad (3a)$$

$$\text{learning: } \theta \leftarrow \theta - \eta_{\theta} \nabla_{\theta} \mathcal{F}_{\text{PC}}(h^*, \theta). \quad (3b)$$

The inference step replaces a standard deep network’s forward and backward passes. The alternating steps are meant to reflect fast (inference) and slow (learning) dynamics of the underlying system.

Finite-inference PC. As written, the argmin in (3a) globally depends on all layer activations. However, gradient descent on this argmin decouples into layer-local updates:

$$h_i(t+1) = h_i(t) - \eta_h \nabla_{h_i} \mathcal{F}_{\text{PC}}(h(t), \theta), \quad i = 1, \dots, L-1, \quad t = 0, \dots, T-1. \quad (4)$$

Each h_i appears in only two terms of $\mathcal{F}_{\text{PC}}$, so $\nabla_{h_i} \mathcal{F}_{\text{PC}}$ depends only on $h_{i-1}(t)$, $h_i(t)$, $h_{i+1}(t)$ and the adjacent weights W_i, W_{i+1} . Activations are typically initialized from their forward-pass values. In general, the number of inference steps required to reach convergence depends on the architecture [Alonso et al., 2024]; in practice, the iteration is typically run either to numerical convergence [Innocenti et al., 2024] or for a fixed T -step budget [Innocenti et al., 2025, Pinchetti et al., 2025]. We refer to this as finite-inference PC to distinguish it from the idealized equilibrated case.

Augmented Lagrangian. Attaching multipliers $\lambda = \{\lambda_i\}_{i=1}^{L-1}$ to the constraints of (1) and keeping the quadratic penalty gives the augmented Lagrangian [Hestenes, 1969, Powell, 1969, Bertsekas, 1976]:

$$\mathcal{L}_{\rho}(h, \theta, \lambda) = \frac{1}{2} \|y - W_L h_{L-1}\|^2 + \sum_{i=1}^{L-1} \lambda_i^{\top} (h_i - \sigma(W_i h_{i-1})) + \frac{\rho}{2} \sum_{i=1}^{L-1} \|h_i - \sigma(W_i h_{i-1})\|^2. \quad (5)$$

Pinning $\lambda \equiv 0$ recovers the PC energy: $\mathcal{F}_{\text{PC}}(h, \theta) = \mathcal{L}_{\rho}(h, \theta, 0)$. Setting $\rho = 0$ recovers the standard Lagrangian of (1).

Method of multipliers. Returning to the constrained optimization problem in (1), we can solve for the activations h by performing dual ascent on the AL (the method of multipliers [Hestenes, 1969,

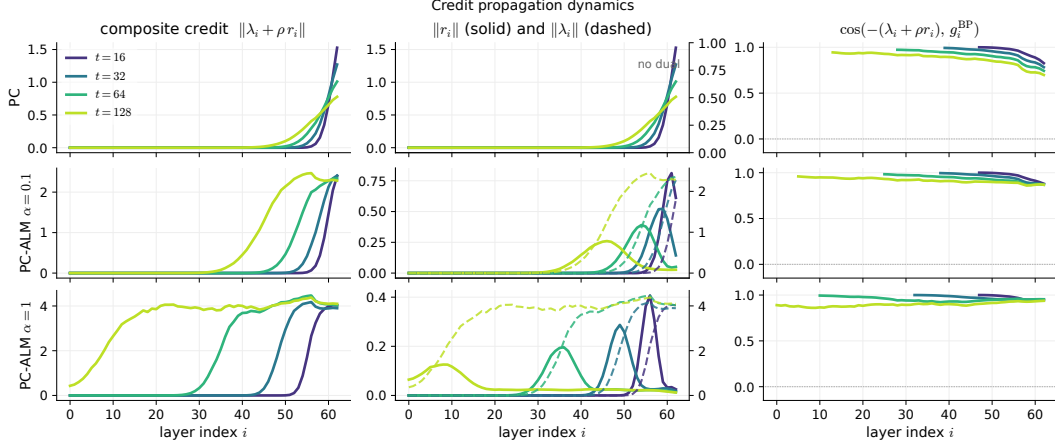


Figure 3: Diffusive versus ballistic credit propagation. ReLU residual MLP with $N = 32$, $L = 64$ on a Fashion-MNIST sample at initialization. The middle plot shows that the prediction error r_i (left axis scale) leads the dual λ_i (right axis scale) across the network. PC-ALM enables faster and more even credit propagation across the network compared to PC.

Powell, 1969, Bertsekas, 1976]). For each sample, initialize $\lambda_i \equiv 0$ and alternate between primal and dual updates

$$\text{primal: } h \leftarrow \arg \min_h \mathcal{L}_\rho(h, \theta, \lambda), \quad (6a)$$

$$\text{dual: } \lambda_i \leftarrow \lambda_i + \rho (h_i - \sigma(W_i h_{i-1})), \quad i = 1, \dots, L-1, \quad (6b)$$

for $k = 1, \dots, K$ outer rounds or until convergence criteria are met. The method of multipliers here is being used to solve the “inference” step but on the *constrained* problem (1), not PC’s quadratic relaxation (2). Therefore, at convergence for settled states (h^*, λ^*) , the activations h^* are simply what one would get from a forward pass. However, the converged duals λ^* represent meaningful local error signals, yielding a weight update:

$$\text{learning: } \theta \leftarrow \theta - \eta_\theta \nabla_\theta \mathcal{L}_\rho(h^*, \theta, \lambda^*), \quad (7)$$

This corresponds exactly to a BP update, as expanded below.

Multipliers as backpropagation gradients. LeCun [1988] observed that at any KKT point of (1), the multipliers of the standard Lagrangian ((5) with $\rho = 0$) coincide with the backpropagation adjoints. With h pinned at the forward-pass values, back-substitution from the output recovers λ in closed form, with λ at the KKT point equal to the BP adjoint at every layer (Appendix A).

Local algorithm via the augmented Lagrangian. This identification suggests an iterative route to the BP adjoints by dual ascent on the standard Lagrangian, which decomposes layerwise (a special case of dual decomposition). On a deep network’s nonconvex Lagrangian, however, the iteration diverges with λ unbounded [Nocedal and Wright, 2006]. The augmented Lagrangian (5) stabilizes the dual update without altering the KKT solution, but at the cost of breaking the layerwise decomposition; the primal argmin (6a) now couples all activations globally. The optimization literature typically resolves this with ADMM [Boyd et al., 2011]. We point out that solving the primal argmin with gradient descent, as in PC, also preserves layer-locality, since $\nabla_h \mathcal{L}_\rho$ at layer i depends only on adjacent layers’ activations and multipliers. While this does not remove the difficulty of reaching a global minimum, the PC-like primal updates paired with dual updates suggest a potential route for layer-local learning with gradients more aligned to BP than PC.

4 Method: PC-ALM

The method of multipliers (6) wraps an outer dual loop around the inner argmin, which must be solved at every outer round. Reaching this inner stationarity is the central bottleneck of PC inference in practice [Pinchetti et al., 2025], and is inherited by MM. We therefore develop PC-ALM, a finite-inference variant of MM that uses the same inference budget T as finite-inference PC. PC-ALM replaces the argmin in (6a) with one gradient descent step on $\mathcal{L}_\rho$, applies the dual update immediately after, and loops $T - 1$ times before a final primal step. The full procedure is Algorithm 1.

Algorithm 1 PC-ALM training

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1: given dataset  $D$ , penalty  $\rho$ , inference steps  $T$ , step sizes  $\eta_h, \alpha, \eta_\theta$ 
2: for each batch  $B \subset D$  do
3:   for each sample  $(x, y) \in B$  in parallel do
4:      $h_0 \leftarrow x$ ;  $h_i \leftarrow \sigma(W_i h_{i-1})$  for  $i = 1, \dots, L - 1$  ▷ forward init
5:      $\lambda_i \leftarrow 0$  for  $i = 1, \dots, L - 1$ 
6:     for  $t = 1, \dots, T - 1$  do
7:        $h_i \leftarrow h_i - \eta_h \nabla_{h_i} \mathcal{L}_\rho(h, \theta, \lambda)$  for  $i = 1, \dots, L - 1$  ▷ primal step
8:        $\lambda_i \leftarrow \lambda_i + \alpha (h_i - \sigma(W_i h_{i-1}))$  for  $i = 1, \dots, L - 1$  ▷ dual step
9:     end for
10:     $h_i \leftarrow h_i - \eta_h \nabla_{h_i} \mathcal{L}_\rho(h, \theta, \lambda)$  for  $i = 1, \dots, L - 1$  ▷ final primal step
11:  end for
12:   $W_i \leftarrow W_i - \eta_\theta \frac{1}{|B|} \sum_{(x,y) \in B} \nabla_{W_i} \mathcal{L}_\rho(h, \theta, \lambda)$  for  $i = 1, \dots, L$  ▷ learning step
13: end for

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Inference gradients. To analyze the difference between $\mathcal{F}_{\text{PC}}$ and $\mathcal{L}_\rho$, we rewrite $\mathcal{L}_\rho$ in a more interpretable form. Completing the square on each cross term in (5),

$$\lambda_i^\top r_i + \frac{\rho}{2} \|r_i\|^2 = \frac{\rho}{2} \left\| h_i - (\sigma(W_i h_{i-1}) - \lambda_i / \rho) \right\|^2 - \frac{1}{2\rho} \|\lambda_i\|^2, \quad (8)$$

where $r_i := h_i - \sigma(W_i h_{i-1})$ denotes the residual (or prediction error) at layer i . Substituting (8) back into (5),

$$\mathcal{L}_\rho(h, \theta, \lambda) = \frac{1}{2} \|y - W_L h_{L-1}\|^2 + \frac{\rho}{2} \sum_{i=1}^{L-1} \left\| h_i - (\sigma(W_i h_{i-1}) - \lambda_i / \rho) \right\|^2 - \frac{1}{2\rho} \sum_{i=1}^{L-1} \|\lambda_i\|^2. \quad (9)$$

The last term has no h -dependence and drops out of $\nabla_h \mathcal{L}_\rho$. The first two terms are the PC free energy (2) with each prediction target shifted from $\sigma(W_i h_{i-1})$ to $\sigma(W_i h_{i-1}) - \lambda_i / \rho$. Differentiating with respect to h_i for $1 \leq i \leq L - 2$,

$$\nabla_{h_i} \mathcal{F}_{\text{PC}} = \rho r_i - W_{i+1}^\top \text{diag}(\sigma'(W_{i+1} h_i)) \rho r_{i+1}, \quad (10)$$

$$\nabla_{h_i} \mathcal{L}_\rho = (\rho r_i + \lambda_i) - W_{i+1}^\top \text{diag}(\sigma'(W_{i+1} h_i)) (\rho r_{i+1} + \lambda_{i+1}). \quad (11)$$

In PC-ALM, each prediction target is shifted by $-\lambda_i / \rho$ relative to PC.

Dual step. After each activity step, the multiplier accumulates the residual:

$$\lambda_i \leftarrow \lambda_i + \alpha r_i, \quad i = 1, \dots, L - 1. \quad (12)$$

The classical MM update (6b) uses rate ρ with the activity solved to stationarity. PC-ALM takes a single inner gradient step instead, so r_i carries unsettled primal gradient on top of constraint infeasibility; we use a separate hyperparameter α for the dual rate. $\alpha = 0$ recovers PC; $\alpha = \rho$ with the inner argmin solved recovers MM.

Weight update. After $T - 1$ primal-dual cycles and a final primal step, weights are updated:

$$\theta \leftarrow \theta - \eta_\theta \nabla_\theta \mathcal{L}_\rho(h, \theta, \lambda). \quad (13)$$

The composite signal $\lambda_i + \rho r_i$ at the weight gradient has λ_i from the integrated dual history and ρr_i from the residual at the final h . By (9), this is the PC weight update with each prediction target shifted by $-\lambda_i / \rho$.

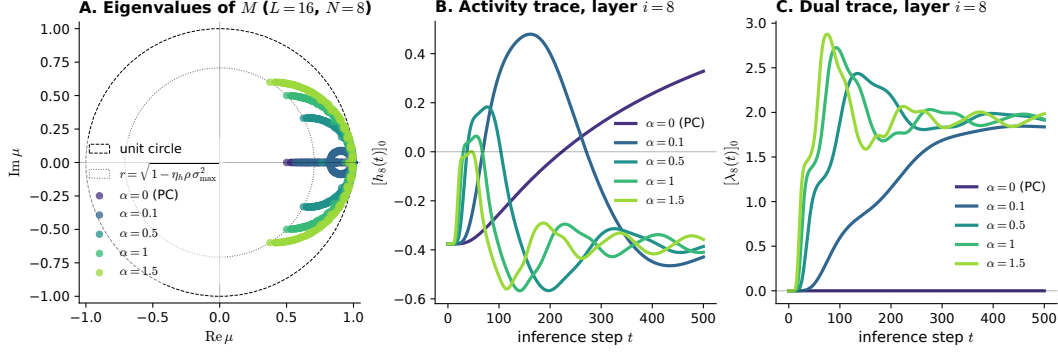


Figure 4: Linear PC-ALM inference dynamics. (A) Eigenvalues of the block iteration matrix M . (B, C) Primal and dual neuron activity traces.

4.1 Interpretation

PC-ALM augments each layer with a per-layer Lagrange multiplier λ_i of the same shape as h_i . On a single sample, the activity loop $t = 0, \dots, T - 1$ initializes h at the forward-pass values and $\lambda \equiv 0$. The first activity step is therefore a standard PC step; with $\lambda \equiv 0$, (11) reduces to (10). The first dual step then injects the layer- i residual r_i into λ_i . From the second activity step onward, the activity gradient at layer i sees the augmented signal $\rho r_i + \lambda_i$ in place of ρr_i alone—a lagged correction entering via the integrated multiplier from prior steps.

At convergence, the activation h_i returns to its forward-pass value (the constraint $h_i = \sigma(W_i h_{i-1})$ is restored, $r_i \rightarrow 0$), while λ_i has integrated to the BP adjoint at that layer (Appendix A). Across the inference loop, λ_i accumulates whatever pressure the layer- i constraint exerts on adjacent activations, and at the fixed point this accumulated pressure is exactly the chain-rule gradient.

Figure 1 illustrates this on a scalar two-layer example. At a fixed weight where equilibrated PC and BP disagree, the dual integrates to the BP adjoint $\lambda_\infty = (y - w_2 h_0) w_2$ while the activation returns to its forward-pass value $h_0 = w_1 x$. The PC-ALM weight update therefore rotates onto BP.

5 Results

We now present a series of theoretical and empirical results for PC-ALM.

5.1 Linear PC-ALM computes exact BP gradients at equilibrium

PC-ALM is an inexact formulation of MM since each primal step is a single gradient step instead of the full argmin. Nevertheless, we show that in a linear PC network, under stability conditions, PC-ALM converges precisely to the KKT point of the constrained optimization problem (Appendix C, Proposition 3). Thus, in linear PCNs, PC-ALM is a layer-local dynamical system that distributes exact BP gradient signals across the network.

5.2 Stability and oscillatory modes

We also provide stability conditions for PC-ALM. The linear PC-ALM dynamics are governed by an iteration matrix M (Equation (17)) and stability requires that its spectral radius be less than 1 (i.e., all eigenvalues inside the open unit disc). Dropping the supervised-loss block ($B = 0$), this reduces to the constraint-only modewise condition

$$\underbrace{\eta_h \sigma_i^2 2\rho}_{\text{PC}} < 4 \quad \underbrace{\eta_h \sigma_i^2 (2\rho + \alpha)}_{\text{PC-ALM}} < 4 \quad (14)$$

at every singular mode σ_i of the constraint operator A (Appendix C, Corollary 1). For $\alpha = 0$ we recover the stability condition for PC. The mean-field parameterizations we use (Appendix B)

stabilize $\sigma_{\max}$ over a wide range of network depths and widths. Setting $\rho = 1$, we roughly obtain $\eta_{\max} \approx 0.5$ and $\alpha_{\max} \approx 2$.

The oscillatory modes of PC-ALM are shown in Figure 4. Notably, in standard PC, all activation dynamics are gradient flows and do not produce oscillatory transients (e.g., eigenvalues on the real line in Figure 4 for $\alpha = 0$). The complex eigenvalues of PC-ALM lie within an annulus with inner radius $\sqrt{1 - \eta_h \rho \sigma_{\max}(A)^2}$ and outer radius 1. The dual rate α slides them along this circle, setting oscillation frequency.

Importantly, convergence to the KKT solution and the BP gradients it provides does not depend on the specific stable choice of η_h and α ; these values only shape the transient dynamics on the path to equilibrium.

5.3 PC-ALM closes the PC-BP gap in deep narrow networks

Beyond the linear analysis, we test PC-ALM experimentally in nonlinear PCNs. We sweep $(N, L) \in \{8, 16, 32, 64, 128\}^2$ in residual MLPs under the mean-field parameterization of Innocenti et al. [2026] (details in Appendix F), comparing PC, PC-ALM, and BP at matched inference budget on Fashion-MNIST and MNIST with identity, tanh, and ReLU activations. PC performance degrades along the depth axis, reproducing the deep-narrow failure mode of Innocenti et al. [2026]. PC-ALM recovers a fraction of the BP gap at a budget of $T = L$ (Figure 10) while closing the gap entirely at a budget of $T = 2L$ across the full range of N , L , and activations (Figure 2). Further experimental details are provided in Appendix F.

5.4 PC-ALM is robust to reparameterization

We investigate whether the performance of PC-ALM is due to the choice of mean-field parameterization, which stabilizes both PC and BP. Holding $(N, L) = (32, 64)$, we sweep two parameterization probes around the mean-field endpoint of Innocenti et al. [2026]: a standard parameterization to mean-field interpolation $\lambda_{\text{sp}} \in \{0, 0.25, 0.5, 0.75, 1\}$ at $\gamma_0 = 1$, and a lazy-to-rich axis $\gamma_0 \in \{0.1, 0.5, 1, 2, 3, 4\}$ at $\lambda_{\text{sp}} = 1$. Across all values, PC-ALM matches the performance of BP (Figs. 6, 7), showing that it remains stable and robust even across unfavorable parameterization regimes.

5.5 Ballistic credit propagation wavefront

In addition to the experimental verification above, we note further surprising dynamical properties of PC-ALM. PCNs are known to exhibit signal decay across layers [Pinchetti et al., 2025]. We reproduce this result as shown in Figure 3 (top). Surprisingly, the primal-dual dynamics of PC-ALM accelerate the credit propagation wavefront across the network until the composite signal $\lambda_i + \rho r_i$ distributes evenly. The prediction-error component r_i propagates ahead of the lagged dual λ_i (Figure 3, middle column). Appendix D gives the linear wavefront calculation. We call this ballistic credit propagation in contrast to PC’s diffusive propagation.

5.6 PC-ALM’s BP alignment exhibits an inflection at a predictable inference step

We observe that PC-ALM provides a modest-but-measurable boost to PC at low inference budgets, but that this boost “snaps” to stronger BP alignment at inference steps of around twice the network depth when using default hyperparameter $\alpha = 1$. This phenomenon is illustrated in Figure 5 (panel A), where the BP alignment yields an inflection point whose location is controlled by α . Higher α pulls the inflection to an earlier inference step. We found this phenomenon to be present for all depths and widths. Further, the location of the inflection can be predicted across a wide range of network depths and widths, namely, at $t \approx L / \sqrt{\alpha \eta_h}$ (Figure 5, panel B), with arguments developed in Appendix D.

5.7 PC-ALM is inference efficient

From a pragmatic standpoint, PC-ALM doubles PC’s activation memory cost, albeit with modest additional compute. Under mean-field scaling, Innocenti et al. [2026] show that increasing width aligns PC’s gradient closer to BP, giving two routes to BP alignment: widening PC, or running

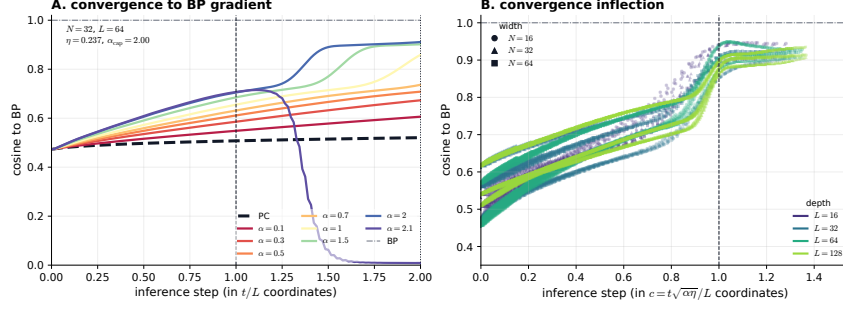


Figure 5: Cosine between the PC-ALM weight gradient and the BP weight gradient at an initialized network. (A) Alignment as a function of iteration, normalized to L . (B) Alignment as a function of iteration t , in units of $c = t\sqrt{\alpha\eta_h}/L$ across different depths and widths.

PC-ALM at fixed width. To achieve a 0.9 cosine similarity to BP, PC-ALM is more efficient by an order of magnitude. Results are in Figure 8, with the argument developed in Appendix E.

6 Discussion

We introduce PC-ALM, a generalization of finite-inference PC that arises as finite-inference dual ascent on the augmented Lagrangian of a feedforward neural network. PC and PC-ALM address two different optimization problems—the quadratic relaxation in PC’s case, and the original constrained problem in PC-ALM’s. They share the same inference budget aside from PC-ALM’s dual update, but exhibit distinct dynamical properties. We verified that PC-ALM is stable for linear PCNs and provides exact BP-aligned credit signals across the network through layer-local accumulation of prediction errors. In nonlinear networks, we further found that PC-ALM is robust and matches BP performance, notably in parameter regimes where PC and BP diverge. We expand on current limitations of our work in Appendix G.

6.1 Connection to other Lagrangian-based methods

PC-ALM is one of several training methods that begin by writing feedforward training as a constrained problem $\min \text{Loss}(h_L, y)$ subject to $h_\ell = \sigma(W_\ell h_{\ell-1})$, and attaching Lagrange multipliers and/or quadratic penalties to the layer constraints. Methods within this family differ in (i) which terms of the augmented Lagrangian they retain, (ii) how they solve the inner activity subproblem, and (iii) how the multipliers, if any, are recovered. Standard PC relies only on the penalty.

Like PC, the BlockProp algorithm [Gotmare et al., 2018] works with a quadratic penalty for a different motivation—model-parallel training across blocks of layers. The authors use an L2 gluing penalty and note that the Taylor-style Lagrange multiplier was unnecessary and less stable in their experiments. They further show that BlockProp with single-layer blocks and a quadratic gluing energy is equivalent to Equilibrium Propagation (EP) of Scellier and Bengio [2017]. EP itself takes a different route around the dual-ascent instability. Rather than iterating the multipliers, it recovers them implicitly by *nudging*—a finite-difference method on the energy gradient between a free fixed point ($\beta = 0$, in EP notation) and a weakly-clamped fixed point ($\beta \rightarrow 0^+$). This approach yields contrastive Hebbian updates that estimate λ^* without ever carrying it as state. EP therefore needs two fixed-point phases; PC-ALM runs primal and dual updates in a single loop. Dual Propagation [Høier et al., 2023] extends the EP nudging idea to a single-phase dyadic-neuron scheme, where the per-neuron difference between nudged states plays a role structurally analogous to PC-ALM’s multiplier. The recent work of Scurria [2026] resembles PC-ALM but via the standard Lagrangian; the similar primal-dual dynamics of their dyadic BP require, however, certain assumptions of nilpotency in the constraint operator equations. We expand on further related work in Appendix H.

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A KKT multipliers as backpropagation adjoints

For a fixed set of weights θ and one sample (x, y) , consider the activity-constrained inner problem induced by (1). At any feasible KKT point of this fixed-weight problem, the Lagrange multipliers satisfy the same reverse recursion as the backpropagation adjoints, and evaluating the weight derivative of the Lagrangian at that point gives the BP weight gradient. We recapitulate this point for the standard Lagrangian (the classical observation of LeCun [1988], recorded here in our notation conventions) and then show that the quadratic augmentation does not change the endpoint.

Throughout, the backpropagation adjoint at hidden layer i is the total derivative of the supervised loss after composing the downstream layers:

$$\delta_i := \frac{d}{dh_i} \frac{1}{2} \|y - W_L h_{L-1}\|^2 \quad \text{evaluated along the forward pass} \quad h_j = \sigma(W_j h_{j-1}), \quad h_0 = x.$$

By the chain rule,

$$\delta_{L-1} = -W_L^\top (y - W_L h_{L-1}), \quad \delta_i = W_{i+1}^\top \text{diag}(\sigma'(W_{i+1} h_i)) \delta_{i+1} \quad \text{for } i = 1, \dots, L-2.$$

A.1 Standard Lagrangian

Setting $\rho = 0$ in (5) gives the standard Lagrangian

$$\mathcal{L}_0(h, \theta, \lambda) = \frac{1}{2} \|y - W_L h_{L-1}\|^2 + \sum_{i=1}^{L-1} \lambda_i^\top (h_i - \sigma(W_i h_{i-1})).$$

Proposition 1. *For fixed θ , at any feasible KKT point of the activity-constrained inner problem, the multipliers of $\mathcal{L}_0$ satisfy $\lambda_i = -\delta_i$ for $i = 1, \dots, L-1$. Consequently, the derivative $\nabla_{W_i} \mathcal{L}_0$ evaluated at this endpoint equals the backpropagation gradient of $\frac{1}{2} \|y - W_L h_{L-1}\|^2$ with respect to W_i .*

Proof. Primal feasibility forces $h_i = \sigma(W_i h_{i-1})$ for $i = 1, \dots, L-1$, so h lies on the forward pass.

Boundary, $i = L-1$: The activation h_{L-1} enters $\mathcal{L}_0$ via the supervised loss and the layer- $(L-1)$ residual:

$$\nabla_{h_{L-1}} \mathcal{L}_0 = -W_L^\top (y - W_L h_{L-1}) + \lambda_{L-1}.$$

Stationarity gives $\lambda_{L-1} = W_L^\top (y - W_L h_{L-1}) = -\delta_{L-1}$.

Recursion, $1 \leq i \leq L-2$: The activation h_i enters the layer- i residual as the LHS (multiplier λ_i) and the layer- $(i+1)$ residual through $\sigma(W_{i+1} h_i)$ (multiplier λ_{i+1}):

$$\nabla_{h_i} \mathcal{L}_0 = \lambda_i - W_{i+1}^\top \text{diag}(\sigma'(W_{i+1} h_i)) \lambda_{i+1}.$$

Stationarity gives $\lambda_i = W_{i+1}^\top \text{diag}(\sigma'(W_{i+1} h_i)) \lambda_{i+1}$. Induction from the boundary yields $\lambda_i = -\delta_i$ for all i .

Weight gradient: For $i = 1, \dots, L-1$, only the i -th layer constraint involves W_i :

$$\nabla_{W_i} \mathcal{L}_0 = -\text{diag}(\sigma'(W_i h_{i-1})) \lambda_i h_{i-1}^\top = \text{diag}(\sigma'(W_i h_{i-1})) \delta_i h_{i-1}^\top,$$

which is the backpropagation gradient at layer i . For the readout, $\nabla_{W_L} \mathcal{L}_0 = -(y - W_L h_{L-1}) h_{L-1}^\top$, the linear-regression gradient. $\square$

The minus sign in $\lambda_i = -\delta_i$ is only a convention: it comes from writing the residual as $r_i = h_i - \sigma(W_i h_{i-1})$. With the opposite residual convention, the multipliers would have the opposite sign. The invariant statement is that the multiplier recursion carries exactly the reverse-mode adjoint information, and the induced weight derivative is the BP gradient.

A.2 Augmented Lagrangian

The augmented Lagrangian (5) adds the quadratic penalty $\frac{\rho}{2} \sum_{i=1}^{L-1} \|r_i\|^2$ to $\mathcal{L}_0$, where $r_i := h_i - \sigma(W_i h_{i-1})$.

Proposition 2. *For the fixed-weight activity problem, the KKT conditions and endpoint weight derivative of $\mathcal{L}_\rho$ coincide with those of $\mathcal{L}_0$. In particular, $\lambda_i = -\delta_i$ at any feasible KKT point, and $\nabla_{W_i} \mathcal{L}_\rho$ evaluated at that endpoint equals the BP weight gradient.*

Proof. The augmentation $\frac{\rho}{2} \sum_i \|r_i\|^2$ is a smooth function of r whose value and all h -, W -derivatives are linear in r and therefore vanish at $r = 0$. Primal feasibility forces $r = 0$, so the stationarity conditions and weight gradient of $\mathcal{L}_\rho$ at any feasible point reduce to those of $\mathcal{L}_0$, and Proposition 1 applies. $\square$

Remark. Proposition 2 is endpoint-only: it characterizes any feasible KKT point but does not say the PC-ALM iteration reaches one. Linear-regime convergence to the KKT point is in Appendix C.

Feedforward graph extension. The same calculation applies to feedforward graphs with skips. At a hidden node, stationarity collects the multiplier on every incoming constraint and subtracts the Jacobian-transpose pullback from every outgoing constraint. On a DAG, solving these equations in reverse topological order gives the usual reverse-mode accumulation. The chain proof above is the single-parent, single-child specialization.

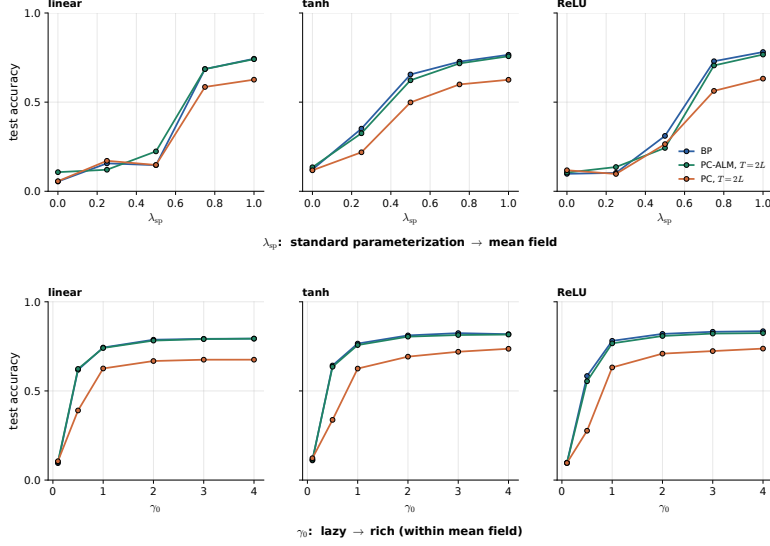


Figure 6: Parameterization sweep on Fashion-MNIST at fixed $(N, L) = (32, 64)$, single seed, $\alpha = 1$. Top row: λ_{sp} interpolates from standard parameterization ($\lambda_{\text{sp}}=0$) to mean field ($\lambda_{\text{sp}}=1$) at $\gamma_0=1$. Bottom row: γ_0 traces the lazy-to-rich axis within the mean-field family at $\lambda_{\text{sp}}=1$. PC-ALM at $T=2L$ tracks BP across both axes; PC at $T=2L$ tracks BP in the lazy regime but lags as the parameterization becomes rich.

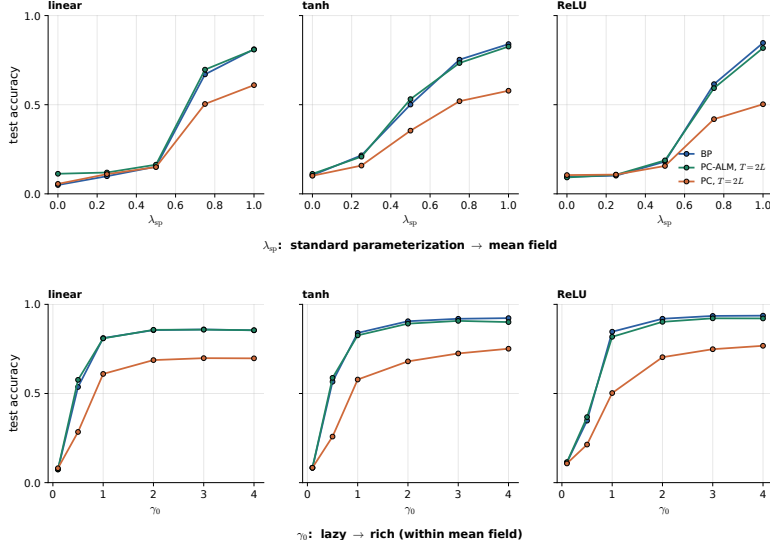


Figure 7: MNIST counterpart of Figure 6. Same axes, same fixed $(N, L) = (32, 64)$, same conclusion: PC-ALM tracks BP across the parameterization sweep; PC lags in the rich regime.

B Architectures

In our main equations we use the layer equations as $h_i = \sigma(W_i h_{i-1})$ for notational simplicity. The experiments use the scaled architectures of Innocenti et al. [2026], where each layer carries a scalar constant that initializes and stabilizes PC at depth and width. The architectures and constants are below.

B.1 Architectures

A depth- L network has weights $W_1, \dots, W_L$ and free hidden states $h_1, \dots, h_{L-1}$ of width N , with $h_0 = x \in \mathbb{R}^D$ clamped and prediction $\hat{y} \in \mathbb{R}^C$. Each layer carries a scalar pre-multiplier a_i .

Chain MLP.

$$h_1 = a_1 W_1 h_0, \quad h_i = a_i W_i \sigma(h_{i-1}) \quad (2 \leq i \leq L-1), \quad \hat{y} = a_L W_L \sigma(h_{L-1}).$$

The input layer is linear; interior and readout layers apply $\sigma \in \{\text{id}, \tanh, \text{ReLU}\}$ before the linear map.

Residual MLP.

$$h_1 = a_1 W_1 h_0, \quad h_i = h_{i-1} + a_i W_i \sigma(h_{i-1}) \quad (2 \leq i \leq L-1), \quad \hat{y} = a_L W_L \sigma(h_{L-1}).$$

Input and readout layers are unchanged from the chain; interior layers add an identity skip from the previous layer. The experiments use this topology.

B.2 Mean-field point

At the mean-field parameterization of [Innocenti et al. \[2026\]](#)—hidden activation exponent $a = 1/2$, init exponent $b = 0$, learning-rate exponent $c = 0$, output-scale exponent $d = 1/2$, residual depth exponent $\alpha_{\text{res}} = 1/2$ (distinct from the PC-ALM dual rate α)—the per-layer pre-multipliers are

$$a_1 = \frac{1}{\sqrt{D}}, \quad a_i = \frac{1}{\sqrt{L N}} \quad (2 \leq i \leq L-1), \quad a_L = \frac{1}{\gamma_0 N},$$

with all weights drawn $\mathcal{N}(0, 1)$ entrywise. The Adam learning rate is coupled to the parameterization [\[Innocenti et al., 2026\]](#),

$$\eta_{\text{Adam}} = \eta_0 \gamma_0^2 \sqrt{N/L},$$

which collapses to the μPC default [\[Innocenti et al., 2025\]](#) at $\gamma_0 = 1$.

C Linear PC-ALM dynamics

This appendix isolates Algorithm 1 in the fixed-weight, linear-activation regime. With $\sigma = \text{id}$ and weights frozen, one PC-ALM inner step is an affine map on the joint activity-dual state (h, λ) . We give the block iteration matrix M , prove convergence to the KKT point under the spectral-radius condition $\text{spr}(M) < 1$, and recover the per-mode Jury bound as a corollary. Architecture-specific information (chain vs. skip MLP) enters only through the block constraint operator A ; the explicit forms are in Appendix B.

C.1 Setup

Take the constrained problem (1) with $\sigma = \text{id}$ and weights θ fixed. The free hidden states are $h = \{h_i\}_{i=1}^{L-1}$, and the multipliers are $\lambda = \{\lambda_i\}_{i=1}^{L-1}$. The residuals can be stacked as a column vector,

$$r(h) = \begin{bmatrix} r_1 \\ \vdots \\ r_{L-1} \end{bmatrix}, \quad r_i = h_i - W_i h_{i-1},$$

The residuals are affine in h and we write $r(h) = Ah + b$, where A is block lower-bidiagonal with I on the diagonal and the negative per-layer linear map below; b absorbs the clamped-input contribution from $h_0 = x$. For a plain chain,

$$A_{\text{chain}} = \begin{bmatrix} I & & & & \\ -W_2 & I & & & \\ & -W_3 & I & & \\ & & \ddots & \ddots & \\ & & & -W_{L-1} & I \end{bmatrix}, \quad b_{\text{chain}} = \begin{bmatrix} -W_1 x \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

The skip form has the same block pattern, with each sub-diagonal block replaced by the negative skip-layer prediction map; the scaled architecture forms are listed in Appendix B.

The supervised loss touches only h_{L-1} :

$$\ell(h) = \frac{1}{2} \|y - Ch\|^2, \quad C = [0 \cdots 0 \ W_L], \quad B := C^\top C.$$

We keep this block separate from the constraint operator: multipliers attach to the network equations $Ah + b = 0$, not to the supervised residual $y - Ch$. The supervised term enters the activity curvature through B and therefore affects the full stability condition below. The augmented Lagrangian (5) in this regime is

$$\mathcal{L}_\rho(h, \lambda) = \frac{1}{2} \|y - Ch\|^2 + \lambda^\top (Ah + b) + \frac{\rho}{2} \|Ah + b\|^2.$$

C.2 One PC-ALM step

Algorithm 1 writes the inference loop as in-place assignments. For the dynamical-systems analysis here, t indexes the repeated inference step, and $h(t), \lambda(t)$ denote the full stacked hidden-state and multiplier vectors before step t . Unrolling one in-place primal-dual cycle gives

$$h(t+1) = h(t) - \eta_h [-C^\top (y - Ch(t)) + A^\top \lambda(t) + \rho A^\top (Ah(t) + b)], \quad (15)$$

$$\lambda(t+1) = \lambda(t) + \alpha (Ah(t+1) + b). \quad (16)$$

Both updates are affine in (h, λ) , so PC-ALM is an affine dynamical system on the joint state.

C.3 Block iteration matrix

Suppose (15)–(16) has a fixed point (h^*, λ^*) . In error coordinates $E(t) := h(t) - h^*$ and $R(t) := \lambda(t) - \lambda^*$, the affine offsets cancel, leaving the linear recursion

$$\begin{bmatrix} E(t+1) \\ R(t+1) \end{bmatrix} = M \begin{bmatrix} E(t) \\ R(t) \end{bmatrix}, \quad M = \begin{bmatrix} I - \eta_h (B + \rho A^\top A) & -\eta_h A^\top \\ \alpha A (I - \eta_h (B + \rho A^\top A)) & I - \alpha \eta_h A A^\top \end{bmatrix}. \quad (17)$$

Chain and skip MLPs share this expression; only A changes (Appendix B).

C.4 Main result: convergence to the KKT point and BP gradient

Proposition 3 (Linear convergence of PC-ALM). *Fix weights θ and take $\sigma = \text{id}$, $\rho > 0$, $\alpha > 0$, $\eta_h > 0$. If $\text{spr}(M) < 1$, then for any initialization $(h(0), \lambda(0))$ the iteration (15)–(16) converges linearly to a unique fixed point (h^*, λ^*) , and:*

1. h^* satisfies $Ah^* + b = 0$ (the linear forward pass);
2. λ^* are the BP adjoints of $\frac{1}{2} \|y - Ch\|^2$ at the linear forward pass;
3. the weight derivative at the fixed point, $\nabla_\theta \mathcal{L}_\rho(h^*, \theta, \lambda^*)$, equals the BP weight gradient.

Proof. In affine coordinates the iteration takes the form $z(t+1) = Mz(t) + c$ for $z = (h, \lambda)$ and a constant c absorbing the input clamp b and supervised-loss offset. $\text{spr}(M) < 1$ implies $I - M$ is invertible, so the fixed-point equation $z^* = Mz^* + c$ has a unique solution. In error coordinates $E(t), R(t)$ defined above, the affine offset cancels, leaving $z(t) - z^* = M^t(z(0) - z^*)$, which converges to 0 at rate $\text{spr}(M)$.

At the fixed point, the dual update (16) forces $Ah^* + b = 0$ since $\alpha > 0$. The activity stationarity (15) then reduces to $-C^\top (y - Ch^*) + A^\top \lambda^* = 0$. These are the KKT conditions of the linear constrained least-squares problem

$$\min_h \frac{1}{2} \|y - Ch\|^2 \quad \text{s.t.} \quad Ah + b = 0.$$

Appendix A (specialized to $\sigma = \text{id}$, $r_i = h_i - W_i h_{i-1}$) identifies these multipliers as the BP adjoints and the weight derivative at the KKT point as the BP weight gradient. $\square$

C.5 Modewise Jury bound

To get a clean per-mode picture, drop the supervised-loss block ($B = 0$); only the constraint penalty remains in M . With the SVD $A = U\Sigma V^\top$ and mode coordinates $x^{(i)}(t) := v_i^\top E(t)$, $y^{(i)}(t) := u_i^\top R(t)$, each positive singular mode evolves independently under the 2×2 map

$$M_i = \begin{bmatrix} 1 - \eta_h \rho \sigma_i^2 & -\eta_h \sigma_i \\ \alpha \sigma_i (1 - \eta_h \rho \sigma_i^2) & 1 - \alpha \eta_h \sigma_i^2 \end{bmatrix}, \quad (18)$$

where σ_i denotes the i -th singular value of A (not the activation σ , which is the identity throughout this appendix).

Corollary 1 (Schur stability per mode). *For $\eta_h, \rho, \alpha > 0$ and $\sigma_i > 0$, M_i is Schur-stable iff*

$$\eta_h \sigma_i^2 (2\rho + \alpha) < 4. \quad (19)$$

Across constraint modes, the bound binds at $\sigma_{\max}(A)$, which sets the activity-step ceiling for the constraint-only iteration.

The eigenvalues of M_i are roots of $\mu^2 - \text{tr}(M_i)\mu + \det(M_i) = 0$ and form a complex conjugate pair when the discriminant $\text{tr}(M_i)^2 - 4\det(M_i)$ is negative; Appendix C.6 works out the precise complex-eigenvalue regime and shows the per-mode response is then a damped oscillation rather than monotone decay. This corollary is not the full stability test when $B \neq 0$: the exact condition remains $\text{spr}(M) < 1$, and readout/output scaling enters through B . In the special case where $B v_i = \nu_i v_i$ is diagonal in the same right singular basis as A , the same Jury calculation gives

$$\eta_h (2\nu_i + (2\rho + \alpha)\sigma_i^2) < 4.$$

Null modes ($\sigma_i = 0$) sit outside the constraint dynamics and are governed by any other curvature in the problem, namely the B block of M .

C.6 Per-mode radius and the role of α

The characteristic polynomial of M_i has trace and determinant

$$\text{tr}(M_i) = 2 - \eta_h \sigma_i^2 (\rho + \alpha), \quad \det(M_i) = 1 - \eta_h \rho \sigma_i^2.$$

Crucially, $\det(M_i)$ does not depend on α ; only the trace does. In the complex-eigenvalue regime $\eta_h \sigma_i^2 (\rho + \alpha)^2 < 4\alpha$, the two roots are conjugate with $|\mu_\pm|^2 = \det(M_i)$, so

$$|\mu_\pm| = \sqrt{1 - \eta_h \rho \sigma_i^2}.$$

For fixed (η_h, ρ, σ_i) , varying α slides the two eigenvalues along the circle of this radius centered at the origin in the complex plane; only the phase changes. Different singular values of A contribute different circles, so the per-mode complex eigenvalues of M populate an annulus

$$\sqrt{1 - \eta_h \rho \sigma_{\max}(A)^2} \leq |\mu| \leq 1$$

in the complex plane. Two consequences:

1. The spectral radius of the complex-mode eigenvalues is set by η_h, ρ , and $\sigma_{\max}(A)$ alone. The dual rate α controls phase (oscillation frequency) but not the magnitude of decay. To accelerate contraction one must push $\eta_h \rho \sigma_{\max}^2$ closer to the Jury ceiling, not push α .
2. As α varies through the boundary $\eta_h \sigma_i^2 (\rho + \alpha)^2 = 4\alpha$, eigenvalues collide on the real axis and split into a complex conjugate pair (or vice versa). At $\alpha = 0$ the per-mode block is upper triangular with eigenvalues 1 and $1 - \eta_h \rho \sigma_i^2$ on the real axis; the 1-eigenvalue is the dual block's invariance.

Figure 4 (in the main text) shows this on a small skip MLP at fixed weights: eigenvalues of M populate the annulus the per-mode bound predicts, and the activity and dual traces show monotone descent at $\alpha = 0$ and damped oscillation at $\alpha > 0$.

C.7 Reduction to penalty PC at $\alpha = 0$

At $\alpha = 0$ the dual is pinned, $\lambda(t) \equiv \lambda(0)$, and M_i collapses to the upper-triangular block

$$M_i|_{\alpha=0} = \begin{bmatrix} 1 - \eta_h \rho \sigma_i^2 & -\eta_h \sigma_i \\ 0 & 1 \end{bmatrix}.$$

The dual mode is invariant (eigenvalue 1), so the joint activity–dual map is not Schur-stable. The activity block is gradient descent on the penalty $\frac{\rho}{2} \|Ah + b\|^2$ along v_i , with the standard penalty-PC stability bound $\eta_h \rho \sigma_i^2 < 2$. Taking the $\alpha \downarrow 0$ limit of Corollary 1 recovers the same activity-block inequality.

C.8 Chain vs. skip

The constraint part of the dynamics is written in A and $\sigma_i(A)$; chain and skip MLPs differ through this operator (Appendix B). The supervised readout contributes the separate block B .

Chain. The off-diagonal block of A is $-W_i$. The depth products $W_j W_{j-1} \cdots$ appear in the inverse/Green’s function of this lower-bidiagonal system, so depth mainly shows up through the spectrum and conditioning of A . Large singular values tighten the activity-step ceiling; small singular values create slow residual modes. Both effects contribute to the finite- T inference difficulty for deep linear PC.

Skip. The off-diagonal block is $-(I + sW_i)$ with $s = 1/\sqrt{LN}$. This makes A_{skip} a discrete difference operator plus a small random perturbation. Under bounded weights, $\sigma_{\max}(A_{\text{skip}})$ remains controlled, so the Jury ceiling on η_h need not shrink just because L grows. Depth can still appear through small modes and conditioning, but the residual parameterization removes the most direct step-size collapse.

D PC-ALM credit-wave propagation

In PC-ALM, each hidden-layer weight update uses the composite credit

$$e_i^t = \lambda_i^t + \rho r_i^t,$$

whereas PC uses only ρr_i^t . At the feasible KKT point the residual vanishes and $e_i^t \rightarrow \lambda_i^t$, the BP adjoint (Appendix A). The finite- T question is how quickly e_i^t acquires useful magnitude and BP alignment at layers far from the readout.

Mode dynamics. In the fixed-weight linear regime of Appendix C with the supervised block B dropped, set $q := \eta_h \sigma^2$ for one singular mode of the constraint operator A . The PC-ALM mode polynomial is

$$\mu^2 - [2 - q(\rho + \alpha)]\mu + (1 - \rho q) = 0, \quad (20)$$

stable when $q(2\rho + \alpha) < 4$ (Corollary 1). In the complex-eigenvalue regime the two roots are $\mu_{\pm} = r e^{\pm i\theta}$, with

$$r = \sqrt{1 - \rho \eta_h \sigma^2}, \quad \cos \theta = \frac{2 - \eta_h \sigma^2(\rho + \alpha)}{2\sqrt{1 - \rho \eta_h \sigma^2}}.$$

The magnitude depends on (ρ, η_h, σ) alone; α enters only through the phase (Appendix C.6). For low-frequency modes, $q \ll 1$ and

$$r \approx \exp\left(-\frac{1}{2} \rho \eta_h \sigma^2\right), \quad \theta \approx \sqrt{\alpha \eta_h} \sigma. \quad (21)$$

Skip-MLP dispersion. Under residual mean-field scaling, A is a layerwise first-difference operator plus a controlled random perturbation (Appendix B). Its long-wavelength singular values follow $\sigma(k) \approx 2 \sin(k/2) \approx k$, with k the layer frequency. Substituting into (21) gives the low- k dispersion relation

$$\mu_{\pm}(k) \approx \exp\left(-\frac{1}{2} \rho \eta_h k^2 \pm i \sqrt{\alpha \eta_h} k\right). \quad (22)$$

The real exponent is diffusive damping; the imaginary exponent is the phase of a damped wave with group velocity

$$v_{\text{credit}} = \sqrt{\alpha \eta_h} \quad \text{layers per inference step.}$$

PC has no phase term: $\mu_{\text{PC}}(k) \approx \exp(-\rho \eta_h k^2)$ is pure heat diffusion. After T inner steps, an output-side credit impulse therefore reaches

$$R_{\text{PC}}(T) = O\left(\sqrt{\rho \eta_h T}\right), \quad R_{\text{ALM}}(T) \approx T\sqrt{\alpha \eta_h}, \quad (23)$$

the latter up to a diffusive front width $O(\sqrt{\rho \eta_h T})$. The strict layerwise “light cone” is one layer per inner step in both cases; (23) measures the depth at which the local credit has crossed a usable amplitude and BP alignment, not where it first arrives.

Predicted inflection step. Equating $R_{\text{ALM}}(T)$ with the network depth L gives the inflection observed in Section 5.6,

$$t_{\text{infl}} \approx \frac{L}{\sqrt{\alpha \eta_h}}, \quad \alpha_{\text{reach}} \approx \frac{L^2}{\eta_h T^2}. \quad (24)$$

At the conditioning step $\eta_h = 1/\lambda_{\text{max}}$ with $\lambda_{\text{max}} = \sigma_{\text{max}}^2(A) \approx 4$ under residual mean-field, $\alpha_{\text{reach}} \approx 4(L/T)^2$. For $L = 64$, $T = 2L$ this gives $\alpha_{\text{reach}} \approx 1$, consistent with the inflection observed near $t \approx 2L$ at $\alpha = 1$ in the headline cell. The Jury bound (19) caps $\alpha < 4/(\eta_h \lambda_{\text{max}}) - 2\rho$; at $\eta_h = 1/\lambda_{\text{max}}$, $\rho = 1$ this is $\alpha < 2$, so the fastest stable reach in T inner steps is $T\sqrt{2/\lambda_{\text{max}}}$. Cells with L exceeding this remain budget-limited even at the cap.

E Complexity comparison

Two routes to BP-aligned gradients. Under the mean-field parameterization of Innocenti et al. [2026], the equilibrated PC energy rescaling satisfies $s(\theta) = 1 + \Theta(L/N)$ (Cor. 4.2; Eq. 19), so the architectural PC–BP mismatch scales as $\Theta(L/N)$ at activity equilibrium. Two routes therefore reduce the PC–BP mismatch: widening the network so that L/N is small, or running PC-ALM at fixed width. These cost, respectively, a width budget tied to depth and an adaptive auxiliary state.

A simplifying assumption on PC’s inner budget. We compare the two routes at a target tolerance ε on the gradient mismatch. PC’s gradient mismatch has two sources: an architectural mismatch $s(\theta) - 1 = \Theta(L/N)$ that persists even at perfect activity equilibrium, and a finite-inference mismatch from a finite inference budget and incomplete equilibration. We assume PC runs at a fixed activity budget $T_{\text{PC}} = \Theta(1)$, independent of ε , large enough that the finite-inference mismatch is negligible beside the architectural term, so PC’s only ε -dependence is its width N_w . We make no matching assumption for PC-ALM, whose inner count remains ε -dependent; the accounting thus favors PC.

Resource comparison. Per training sample, PC at width N_w runs T_{PC} activity-update steps (fixed, by the assumption above); PC-ALM at fixed width N_0 runs K_ε joint primal–dual iterations, in the linear fixed-weight setting of Proposition 3, to drive the mismatch below ε . The PC-ALM convergence rate used below is the linear fixed-weight rate of Proposition 3; the nonlinear ReLU experiment in Figure 8 is an empirical diagnostic rather than a consequence of this asymptotic argument.

The PC-ALM inner count follows from linear convergence. By Proposition 3, the joint primal–dual iteration contracts at rate $\text{spr}(M) < 1$, where $\text{spr}(\cdot)$ denotes the spectral radius. Driving the gradient mismatch below ε therefore requires

$$K_\varepsilon = O\left(\frac{\log(\varepsilon)}{\log(\text{spr}(M))}\right)$$

joint primal–dual iterations. For PC there is no auxiliary state, and by the fixed-budget assumption no ε -dependent inner count. Its gradient mismatch is set by the architectural rescaling $s(\theta) - 1 = \Theta(L/N)$ [Innocenti et al., 2026, Eq. 19], so ε is controlled by the choice of width, $N_w(\varepsilon) = \Theta(L/\varepsilon)$. Widening pays for tighter ε linearly in width, while PC-ALM pays for tighter ε logarithmically in joint iterations. The dominant costs are summarized in Table 1. Symbol guide: L is depth, B is batch size, N_0 is PC-ALM’s fixed width, N_w is PC’s width, K_ε is the number of joint PC-ALM primal–dual iterations to reach tolerance ε , T_{PC} is PC’s fixed (ε -independent) activity-inference budget, λ denotes the ALM multiplier state, and M is the linear PC-ALM iteration matrix from Proposition 3.

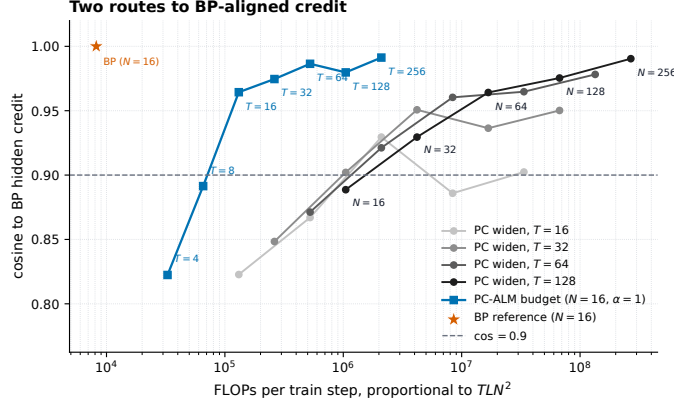


Figure 8: Two routes to BP-aligned hidden credit at $L = 32$, ReLU, Fashion-MNIST. Here T denotes the plotted per-sample update budget: activity steps for PC and joint primal-dual cycles for PC-ALM. Gray: PC at fixed inference budget T , sweeping hidden width N . Blue: PC-ALM ($\alpha = 1$) at fixed $N = 16$, sweeping T . Orange star: BP reference at $N = 16$. The horizontal axis is per-step compute, proportional to TLN^2 . In this $L = 32$, ReLU/Fashion-MNIST setting, the PC-ALM branch reaches 0.9 cosine similarity to BP at roughly an order of magnitude less compute than the PC width sweeps shown.

	Parameters	Memory (activations + multipliers)	Total time
PC-ALM (fixed width N_0)	LN_0^2	$2LN_0B$	$K_\varepsilon LN_0^2B$
PC at width N_w	LN_w^2	LN_wB	$T_{PC} LN_w^2B$
Ratio (ALM / PC)	N_0^2/N_w^2	$2N_0/N_w$	$K_\varepsilon N_0^2/(T_{PC}N_w^2)$

Table 1: Resource costs for PC-ALM at fixed width N_0 and PC at width N_w , both under mean-field scaling at matched gradient tolerance ε , with PC’s inner budget T_{PC} held fixed and ε -independent.

Space asymptotics. Substituting $N_w = \Theta(L/\varepsilon)$ into the bottom row of Table 1 gives the asymptotic ratios. Parameters scale as $\Theta(N_0^2\varepsilon^2/L^2)$ and activation memory as $\Theta(N_0\varepsilon/L)$. Both are pure architectural ratios, free of the inner-iteration counts; PC-ALM dominates PC on space whenever $N_0\varepsilon \ll L$.

Time asymptotics. Substituting $N_w = \Theta(L/\varepsilon)$ and K_ε asymptotics gives

$$\frac{\text{time}_{\text{ALM}}}{\text{time}_{\text{PC}}} = O\left(\frac{N_0^2\varepsilon^2}{L^2} \cdot \frac{\log(\varepsilon)}{T_{PC} \log(\text{spr}(M))}\right), \quad (25)$$

for fixed L , N_0 and a stable PC-ALM iteration matrix M with $\text{spr}(M) < 1$. Under the fixed-budget assumption $T_{PC} = \Theta(1)$, the ratio is $O(|\varepsilon^2 \log(\varepsilon)|)$. The architectural factor $N_0^2\varepsilon^2/L^2$ drives the entire ratio to zero as ε tightens, provided the PC-ALM spectral gap is bounded away from zero, since $\varepsilon^2 \log(1/\varepsilon) \rightarrow 0$. PC-ALM therefore dominates PC on time in the small- ε limit. If $\text{spr}(M) \rightarrow 1$ with depth, weights, or conditioning, the constant can degrade. A symmetric treatment of PC’s inner budget, charging PC an ε -dependent inference budget as well, would cancel the $\log(1/\varepsilon)$ factor up to the differing conditioning of the two inference problems.

F Experimental details

For our experiments we closely follow the setup of Innocenti et al. [2026], using their residual MLP architecture (Appendix B) and mean-field parameterization. The parameterization stabilizes $\sigma_{\max}(A)$, so a default η_h would suffice in principle. However, $\sigma_{\max}^2(A)$ is sensitive in shallow networks, and α and η_h are coupled. We therefore compute an empirical $\lambda_{\max} := \sigma_{\max}^2(A)$ for each (N, L) and dataset, and set $\eta_h = 1/\lambda_{\max}$. This places α in an interpretable $[0, 2]$ range. All networks are

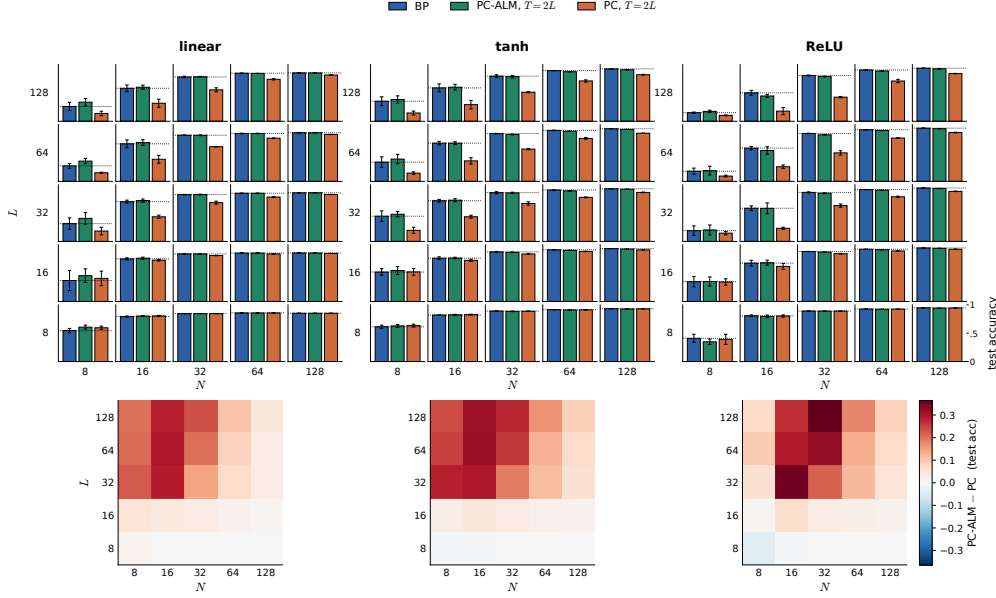


Figure 9: MNIST counterpart of Figure 2. Same (N, L) grid, three seeds, $\alpha = 1$. Same regime structure: PC-ALM at $T = 2L$ tracks BP across the grid; PC drops in deep narrow networks; the gain concentrates in deep, narrow cells.

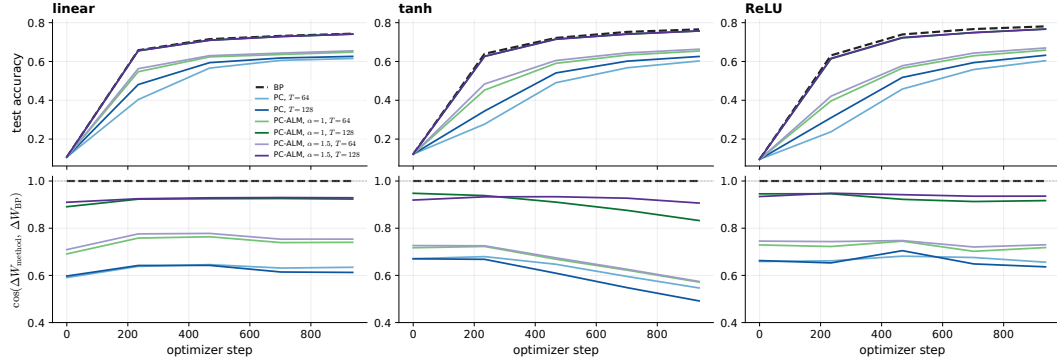


Figure 10: Training curves for one epoch at $N = 32, L = 64$ on Fashion-MNIST across identity / tanh / ReLU.

trained for one epoch on MNIST [LeCun et al., 1998] and Fashion-MNIST [Xiao et al., 2017]. We additionally swept η_h for several PC networks on a tight grid, and observed negligibly improved performance at higher η_h followed by collapse beyond $2/\lambda_{\max}$ as expected. The η_h sweep was unable to improve PC’s performance in any meaningful way compared to PC-ALM, which we use as justification for the default η_h choice. We use a batch size of 64.

G Limitations

Structural restrictions. The convergence proof for PC-ALM (Appendix C, Proposition 3) and the credit-wave analysis of Appendix D are restricted to networks with identity activations and no bias terms. Our experiments use activations (identity, tanh, ReLU) across the full $(N, L) \in \{8, 16, 32, 64, 128\}^2$ grid and PC-ALM closes the PC-BP gap throughout, but a convergence guarantee covering this setting is open. We also do not include bias terms in our implementation, opting for

weights only. This simplification to linear and bias-free networks mirrors the broader PC literature: the saddle analysis of [Innocenti et al. \[2024\]](#) likewise uses linear networks without biases for theoretical development, and nonlinear networks for empirical verification. The μ PC scaling analysis of [Innocenti et al. \[2025\]](#) is likewise linear and bias-free in its theoretical content while running nonlinear (bias-free) experiments. The analysis of [Song et al. \[2024\]](#) relies on linearizing a nonlinear PC network around a feedforward initialization.

Empirical scope. Experiments are restricted to MNIST and Fashion-MNIST, residual MLPs, and one epoch of training. Broader evaluation on larger and more diverse datasets, longer training horizons, and architectures beyond residual MLPs (convolutional, attention-based) remains future work. The width-depth grid covers $(N, L) \in \{8, 16, 32, 64, 128\}^2$, which probes the regime relevant to the PC-BP gap but is small relative to production networks.

Inference budgets. The correspondence between the weight update and the BP gradient (Appendix A) only holds at equilibrium. Such an equilibrium may fail to be reached with a finite inference budget, causing gradient misalignment. Moreover, the inference budget necessary to reach equilibrium (within a tolerance) grows with network depth.

H Additional Related Work

There is a large body of work on biologically plausible, local learning methods. Target propagation [[Bengio, 2014](#)] and difference target propagation [[Lee et al., 2015](#)] have been shown to be related to predictive coding [[Millidge et al., 2022a](#)]. We believe that connections of target propagation to PC-ALM in particular merit investigation. The predictive coding literature often leads with a probabilistic framework [[Rao and Ballard, 1999](#), [Friston and Kiebel, 2009](#), [Millidge et al., 2021](#), [Salvatori et al., 2026](#)]; the framing of PC as a quadratic penalty on the constrained optimization problem is, to our knowledge, less common (though see [Seely \[2025\]](#)). One benefit of the PC formalism is that it can handle non-DAG topologies as studied in [Salvatori et al. \[2022\]](#). The AL framework also extends to non-DAG architectures; at least in a linear PCN, recurrent topologies change the KKT point but do not affect convergence results. With nonlinear activations, the primal-dual dynamics can support nonlinear systems with limit cycles and other nonlinear behavior, which is not possible in standard PC’s gradient-flow inference; we leave this to future investigation. PC has also been cast in terms of sheaf theory, which offers tools for analyzing recurrent topologies [[Seely, 2025](#)]; closely related is the work of [Bosca and Ghrist \[2026\]](#), which extends the sheaf-theoretic framing to ReLU MLPs with resulting PC-like inference and learning steps. From this sheaf-theoretic perspective, the primal-dual dynamics of PC-ALM can be viewed as an application of the homological programming methods of [Hansen and Ghrist \[2019\]](#). Prior work on “PC approximates backprop” is highlighted by [Innocenti et al. \[2026\]](#); earlier work showed that with a “fixed prediction assumption” (a modification of standard PC), error propagation can converge to BP [[Millidge et al., 2022b](#)], albeit in a large number of iterations; this work was further studied in [Rosenbaum \[2022\]](#). Finally, we remind the reader that while PC and PC-ALM address one aspect of BP’s bio-implausibility, they do not address weight transport (the gradient of the energy still uses the transpose of the layer’s weights [[Bredenberg et al., 2023](#)]).